Cyberpsychology, Behavior, and Social Networking
Volume 00, Number 00, 2025

DOI: 10.1089/cyber.2024.0614

Beyond Prompt Engineering: Exploring Collaborative Dialogue with Generative AI for Problem-Solving

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Abstract

Since generative AI (GenAI) launched, interactions between humans and artificial intelligence have rapidly evolved. This study explores how discourse practices in human–AI interactions influence collaborative problem-solving with ChatGPT. Grounded in small group dynamics research, the investigation examines whether three discourse practices (i.e., construction, co-construction, and constructive conflict) help users and ChatGPT find common ground (i.e., shared cognition) to effectively and efficiently complete creative writing projects. Participants completed creative writing projects using ChatGPT for a month and subsequently responded to surveys assessing discourse practices, shared cognition, team effectiveness, and concepts from the Technology Acceptance Model (TAM). Results indicate that discourse practices significantly predict shared cognition, which in turn mediates the relationship between discourse practices and perceived team effectiveness. Furthermore, shared cognition positively influences behavioral intentions to use ChatGPT, primarily through its effects on perceived ease of use and perceived usefulness, aligning with TAM predictions. These findings have important implications for understanding users' communication styles, suggesting that principles from human-to-human interactions can and should be applied to AI conversations. The current study proposes that ChatGPT is capable of mirroring and reciprocating these discourse practices, which may open the possibility to begin optimizing dialogue with GenAI similar to the goals of prompt engineering.

Keywords: collaboration, discourse practices, generative AI, human–AI interaction, technology acceptance model

Introduction

Generative AI (GenAI) is a growing subset of human–AI interaction that trains with large language models to converse and assist people with daily chores, work tasks, and leisure activities[1]. Past research on communication with GenAI has focused on the iterative process of prompt engineering that refines users' queries for optimizing a GenAI's response.[2,3] However, the two-way communication between users and GenAI can also resemble a discursive process where both parties share knowledge, exchange ideas, and form a shared understanding of a topic before mutually constructing new insights and ideas. This parallels knowledge-building practices in human communication that unfold over time such as collaborative learning,[4,5] shared mental models,[6] and distributed cognition[7] in group settings. As such, research must shift toward exploring sophisticated discourse structures to optimize communication with GenAI.

Research on human-to-human communication has identified three crucial discourse practices that increase the perceived performance and effectiveness of problem-solving in groups (i.e., construction, co-construction, and constructive conflict).[5,8] The current study examines whether implementing such discourse practices can foster efficient and effective collaborative discussions with GenAI and increase intentions to use GenAI through the concepts proposed in the technology acceptance model (TAM: ease of use, perceived usefulness [PU], and intention to use).[9]

Prompt engineering and discourse practices with GenAI

The goal of prompt engineering is typically restricted to producing a single or limited set of GenAI responses.[10,12–14] Although research has begun exploring GenAI responses that compound and integrate users' prior input to create a structured dialogue (i.e., multi-turn dialogue),[15,16] this literature similarly focuses on strategies for minimizing the necessary input from users by tweaking instructions or reiterating prior responses. However, prior research on small groups has demonstrated specific practices that shaped dialogue between people into a structured discourse capable of min-maxing group members' effort and effectiveness in problem-solving.[5,17–19] These discourse practices increase the proficiency of communication by building a mutual understanding of a problem (i.e., shared cognition). This typically begins when one member gives meaning to the situation by offering their interpretation

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of the problem and potential solutions (i.e., construction of meaning).[5,20] Subsequently, the meaning offered by the initial explanation is modified and refined by other members who may have different points of view or perspectives (i.e., co-construction of meaning).[5,21,22] Another critical practice occurs when conflicting interpretations are negotiated (i.e., constructive conflict) until convergence on a joint meaning is reached, resulting in shared cognition.[5,23] When this shared cognition is reached through clarifications and argumentation, the group is better equipped to promptly find effective solutions, which typically increases perceptions of team performance.[8,19]

The current study

Research has shown that GenAI can adapt and sufficiently address users' requests with sufficient prompt engineering,[11–16] but it is unclear how users can maximize GenAI performance over prolonged periods of collaborative deliberation and coordinated problem-solving. The current study examines whether discourse practices that are crucial for effective problem-solving in small groups can lead to similar benefits for users' communication with GenAI. Therefore, the current study proposes the following research questions:

> RQ1: Will users who implement discourse practices (i.e., construction, co-construction, and constructive conflict) perceive ChatGPT to share their understanding of a problem and potential solutions (i.e., shared cognition) and, as a result, subsequently perceive their discussion as more effective?

Given the expanding application of GenAI, its capability to perform these tasks may play a critical role in users continued use of the technology. Indeed, previous research on the TAM suggests there are two factors that influence users' behavioral intentions (BI) to adopt and stay committed to new technologies: PU and perceived ease of use (PEU).[9] In the context of GenAI, PU refers to how effectively users believe a GenAI can help them achieve their goals, while PEU describes how easy it is to interact with the GenAI in the pursuit of these goals. The TAM also suggests there are external factors that directly and indirectly (through PEU) influence PU.[9] These external factors can include subjective norms, demographic factors (e.g., age and gender),[24] and technology-specific factors like communication styles with GenAI.[25] As such, the current study examines whether shared cognition with GenAI resulting from discourse practices will directly lead to greater PU and indirectly through increases in PEU. Subsequently, this research question also examines whether PU will then increase users' BI to use and continue using GenAI as predicted by the TAM (see Figure 2).

> RQ2: Will a shared understanding of a problem and potential solutions between users and ChatGPT (i.e., shared cognition) directly and indirectly (through greater PEU) affect PU of the technology? Will then PU increase BI to use and continue using ChatGPT?

Methods

This study was approved by Institutional Review Boards at both Indiana University (protocol #: 19104) and Jeju University (protocol #: 2023-034).

Participants

Participants were 51 university students (23 male, 25 female, 3 unidentified; mean age 21; 100% Asian) enrolled in one of three literature courses offered by the same professor at a national university in South Korea. Participants were over 18, native Korean-speaking, and provided informed consent. No incentives were given. *A priori* G*Power[26] indicated $N = 27$ ($f^2 = 0.33$ based on a conservative estimate from similar past research,[24] $1\text{-}\beta = 0.80$, a significance level of $\alpha = 0.05$ with four predictors). See the full output is available on Open Science Framework (OSF, https://go.iu.edu/8tpp).

Procedure

At the beginning of each course, participants were instructed to complete a creative writing project in collaboration with ChatGPT (e.g., brainstorming, information acquisition, and writing process). Each participant's assignment in the three courses involved: (a) Creating a short SF story with AI as the theme using ChatGPT in the "Understanding Science Fiction" course, (b) Crafting a short story about the impact and changes AI might bring to future education using ChatGPT in the "theory of Korean contemporary literature" course, and (c) engaging in creative post-reading writing using ChatGPT in the "theory of reading pedagogy" course, such as writing letters to authors or creating works related to the themes of books.

Although the final output for each project differed, the parameters (e.g., assignment timeline and minimum length of 1,000 words) and guidelines for the creative writing assignments were the same in all three courses. To ensure uniformity and balanced usage skills, the instructor demonstrated ChatGPT3.5 in all courses, highlighting basic prompt engineering and demonstrating best practices for utilizing GenAI in creative writing exercises (see Table 1). Out of the participants, more than half ($N = 28$) used version 3.5, followed by version 4.0 ($N = 20$), version 3.0 ($N = 1$), and not answered ($N = 2$). All participants answered survey questions after engaging with GenAI for a month. A month timeline was deemed necessary by the instructor's schedule and teaching practices, which is corroborated by prior research suggesting such creative writing interventions should be employed over 4–8 weeks.[27,28] Additionally, this timeframe provided participants with substantial time to explore and solidify perceptions of discourse practices with GenAI.

Measures

Participants responded to all questions on a 7-point Likert-type scale (1 = Strongly Disagree, 7 = Strongly Agree). See Table 2 for example items and descriptive statistics for all measures.

Discourse practices. Participants' perceptions of their own and ChatGPT's participation in discourse practices were assessed using the average of three subscales adapted from the Team Learning Beliefs and Behaviors Questionnaire (TLBBQ)[5]: construction, constructive conflict, and co-construction.

Table 1. Participants' Final Output Guideline

Common guideline	
Task duration: 4 weeks (between May and June 2023)	
Words limits minimum 1,000 words	
Collaborate with ChatGPT throughout the assignment. Utilize ChatGPT for brainstorming and gathering information, as well as during the creative writing process. For example, you can ask about various elements of the novel, request content generation, collaborate on character development, seek information on unfamiliar words, establish characters' relationships for developing the story, and more.	
Assignments guideline	
Course 1.	Create a science fiction short novel focused on artificial intelligence.
Course 2.	Compose a short story exploring the potential impact and changes AI might bring to future education.
Course 3.	Produce a reading reflection on a nonfiction book of your choice. The output can take various forms of writing, such as reflective poetry, fictional stories, essays, letters to the author, or other creative formats.

Note: Course 1 = Understanding SF (Science Fiction), Course 2 = The theory of Korean contemporary literature, Course 3 = Theory of reading pedagogy.

Shared Cognition. Participants' perceptions of a common understanding between themselves and ChatGPT of the assignment and how to complete the assignment were measured using two items from the TLBBQ.[5]

TAM. Participants' evaluations of technology acceptance with GenAI were measured using three core variables from subscales of the TAM[9]: PEU, PU, and BIs.

Team Effectiveness. Participants' perceptions of team effectiveness while collaborating with ChatGPT were measured using four items from the TLBBQ.[5]

Results

The macro PROCESS version 4.2[29] for SPSS was used to conduct a simple mediation analysis (Model 4) for the first research question and a serial mediation analysis (Model 6) for the second research question. Both analyses utilized 5,000 bootstrap samples and 95% bias-corrected confidence intervals.[30] See Table 3 and Figures 1 and 2 for all individual paths.

RQ1. Discourse practices, shared cognition, and team effectiveness

As predicted, the indirect effect of discourse practices on team effectiveness through shared cognition was statistically significant ($\beta = 0.38$, standard error [SE] = 0.11, 95% confidence interval [CI] = 0.18–0.63), and the direct effect between discourse practice and team effectiveness was not significant when controlling for shared cognition ($\beta = 0.16$, SE = 0.11, $t = 1.47$, $p = 0.14$, 95% CI = –0.06 to 0.39).

RQ2. Shared cognition and TAM

Consistent with TAM, the analysis indicated that shared cognition indirectly affected BI through PU ($\beta = 0.27$, SE = 0.13, 95% CI = 0.02–0.53). Furthermore, in line with prior research demonstrating a serial relationship from PEU to

Table 2. Measurements

Variable		*M*	*SD*	*Cronbach's α*
DP	Construction (3 items) (e.g., "I shared all relevant information and ideas with ChatGPT")	5.33	1.00	0.88
	Constructive conflict (3 items) (e.g., "ChatGPT and I handled differences by addressing them directly")			
	Co-construction (3 items) (e.g., "Information from ChatGPT was complemented with my information")			
SC	(2 items: e.g., "ChatGPT and I had a common understanding of the assignment we had to handle")	5.58	1.03	0.90
TE	(4 items: e.g., "collaborating with ChatGPT as a team:We have completed the assignment in a way we all agree upon")	5.55	.90	0.86
TAM	Perceived usefulness (10 items) ("ChatGPT increased my productivity")	5.87	0.84	0.93
	Perceived ease of use (10 items) ("I believe that it was easy to get ChatGPT to do what I want it to do")	4.97	0.73	0.80
	Behavioral intention of use (3 items) ("I plan to use ChatGPT as part of my regular work")	5.30	1.08	0.87

Note: Participants indicated on the 7-point Likert scale how much they agree with the following statements about the measurement.
DP, Discourse Practice; SC, Shared Cognition; TE, Team Effectiveness; TAM, Technology Acceptance Model.

Table 3. Direct, Indirect, and Total Effects of Sequential Mediation Analysis

Variables	β	*SE*	*t*	p	*95% CI*
Direct effects					
SC→PEU	0.34	0.08	3.86	***	[0.16, 0.52]
SC → PU	0.43	0.09	4.65	***	[0.24, 0.62]
PEU → PU	0.35	0.13	2.70	**	[0.09, 0.62]
SC → BI	0.12	0.18	0.67	0.50	[–0.25, 0.50]
PEU → BI	–0.22	0.23	–0.93	0.35	[–0.69, 0.25]
PU → BI	0.63	0.24	2.61	*	[0.14, 1.11]
Indirect effects					
SC→PEU→BI	–0.07	0.07			[–0.25, 0.09]
SC → PU →BI	0.27	0.13			[0.02, 0.53]
SC → PEU → PU → BI	0.07	0.05			[0.00, 0.20]
Total effects					
SC → BI	0.40	0.14	2.83	**	[0.11, 0.68]

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
SC, Shared Cognition; PU, Perceived Usefulness; PEU, Perceived Ease of Use; BI, Behavioral Intention; CI, confidence interval.

PU, it also influenced BI through both mediators in sequence ($\beta = 0.07$, SE = 0.05, 95% CI = 0.00–0.20).

Discussion

Much of the literature on communication with GenAI has primarily occurred in the field of computer science and health, with limited insights from disciplines related to effective and efficient discourse.[31] Research in the social sciences has long studied the dynamics of small groups including which discourse practices create the most effective collaborations between group members.[5,17–21] The current study suggests that GenAI can participate in such discourse practices when implemented by users, and that such collaboration increases users' positive beliefs about GenAI and intentions to use GenAI.

Results for the first research question demonstrated that using optimal discourse practices led participants to believe ChatGPT not only understood the discussed problem, but that ChatGPT had a shared understanding with the user on how to "handle" it (i.e., shared cognition). In line with prior research on small groups,[5,6] this shared cognition between users and ChatGPT resulted in greater perceptions of discussion effectiveness and satisfaction. As such, the findings suggest that users seeking to optimize extended problem-solving discussions should capitalize on ChatGPT's ability to participate in and follow the structured discourse that results from construction, co-construction, and constructive conflict practices. Indeed, in the current study, participants reported ChatGPT's adoption of these practices in addition to their own implementation (e.g., "ChatGPT and I tended to handle differences of opinions by addressing them directly," constructive conflict). This suggests that ChatGPT was capable of mirroring and reciprocating these discourse practices and that users' experiences can benefit from the well-established knowledge of human-to-human communication and dynamics.

The second research question highlighted this potential impact of discourse practices on users' experiences with ChatGPT. Concepts from the TAM[9] were used to determine whether the shared cognition that results from the successful execution of these discourse practices affected users' interactions with and perceptions of ChatGPT as a useful tool for completing their project. Specifically, the results followed the predictions of the TAM by indicating that a shared

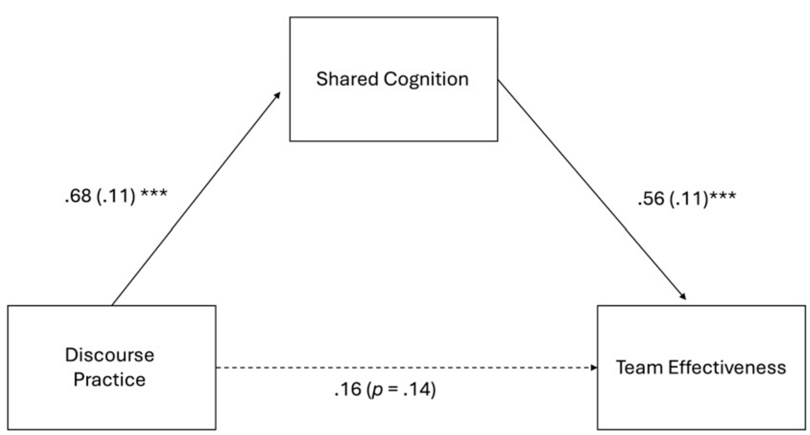

FIG. 1. A path diagram of the simple mediation model (process model 4). Note: Values are unstandardized point estimates for each direct path, and their standard errors in parentheses. Significant path estimates are denoted as bold lines. $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$. Results indicated that the model explained 43% of the variance in shared cognition ($R^2 = 0.43$) and 59% of the variance in team effectiveness ($R^2 = 0.59$).

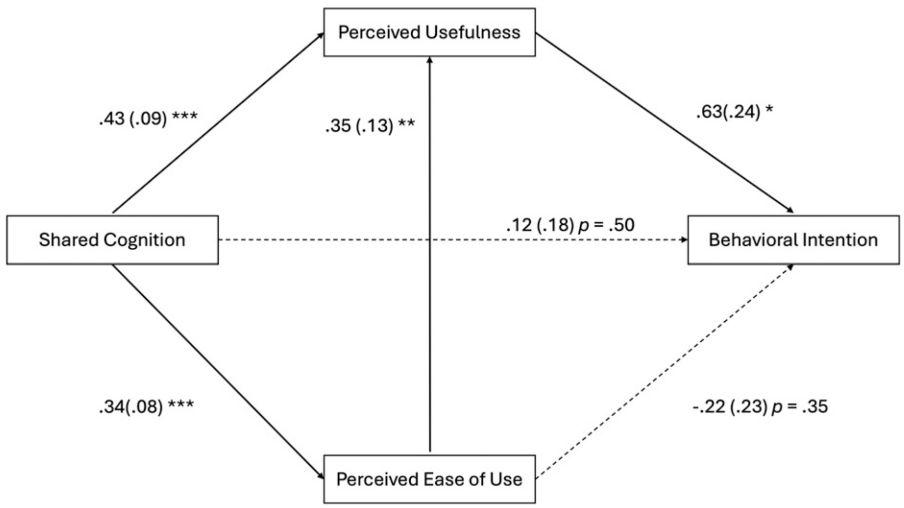

FIG. 2. A path diagram of the sequential mediation model (process model 6). Note: Values are unstandardized point estimates for each direct path and their standard errors in parentheses. Significant path estimates are denoted as bold lines. $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$. Results indicated that the model explained 23% of the variance in PEU ($R^2 = 0.23$), 53% of the variance in PU ($R^2 = 0.53$), and 25% of the variance in BI ($R^2 = 0.25$). BI, behavioral intention; PEU, perceived ease of use; PU, perceived usefulness.

understanding of the problem with ChatGPT (i.e., an external factor) first increased perceptions of the technology as easy to use (PEU) which then increased its PU and, as a result, increased intentions to frequently use ChatGPT as part of their future work (BI). As such, these discourse practices and the shared cognition they produce seem to simplify and expedite collaborative discussions with GenAI, much in the same way they work for small group communication between people.

Limitations

The current study is limited in its ability to assess how and when discourse practices were initiated and developed in participants' discussions with ChatGPT. Transcripts of participants' discussions with ChatGPT were not recorded for analysis. Therefore, it is unclear whether these practices were intentionally provoked by participants or if they were an unintended product of participants' unique communication styles and ChatGPT's specific responses. Future experiments can educate and encourage users to implement discourse practices along with varying the instructions users provide ChatGPT for structuring their discussions. Methodologically, the current study prioritized ethical considerations and participant autonomy by employing minimally invasive survey methods rather than comprehensive data logging, though future research could balance privacy concerns with richer analysis via partial opt-in transcript sampling or automated anonymization.

In addition, participants' variation in model use (ChatGPT 3.5 vs. 4.0) may have affected participants' experiences with GenAI. The current study found that the result did not change with GPT version as a covariate, so we reported the simpler analysis. The analyses with covariates can be found at OSF (https://go.iu.edu/8tpp). While the capabilities of the different models for discursive processes are currently unknown, the superior reasoning, contextual understanding, and output quality of 4.0 compared with 3.5 could have advantaged some participants.

Conclusion

The current study suggests that discourse practices can shape users' and ChatGPT's joint understanding of a problem which fosters effective problem-solving and establishes ChatGPT as a valuable and key resource for the future The ability of ChatGPT to conform and adhere to discourse practices may open the possibility to begin optimizing dialogue with GenAI (i.e., discourse engineering) similar to how prompt engineering optimizes GenAI's responses to specific questions and instructions.[4,5,32] Indeed, simply educating users about effective discourse practices traditionally found in human communication may maximize their usage of GenAI over time with more difficult or nuanced tasks. This suggests future research should further explore GenAI's aptitude for other sophisticated communication phenomena. For example, in educational contexts, instructors can operationalize discourse engineering by crafting AI prompts that follow Socratic questioning and reflective-inquiry patterns, guiding students through metacognitive dialogues that foster critical thinking and self-regulation.[33,34] The richer interactions resulting from such discourse engineering may also elevate GenAI's standing as collaborators who can offer users meaningful and satisfying work experiences.[35] Future research should further explore the concept of discourse engineering as a potentially fruitful avenue for scholarly inquiry and GenAI development.

Authors' Contributions

S.H.L.: Conceptualization, methodology, investigation, formal analysis, writing—original draft, and writing—review and editing. J.A.V.: Conceptualization, methodology, formal analysis, writing—original draft, and writing—review and editing. D.-w.N.: Conceptualization and investigation.

Author Disclosure Statement

The authors have no conflicts of interest to disclose.

Funding Information

No funding was received for conducting this study.

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